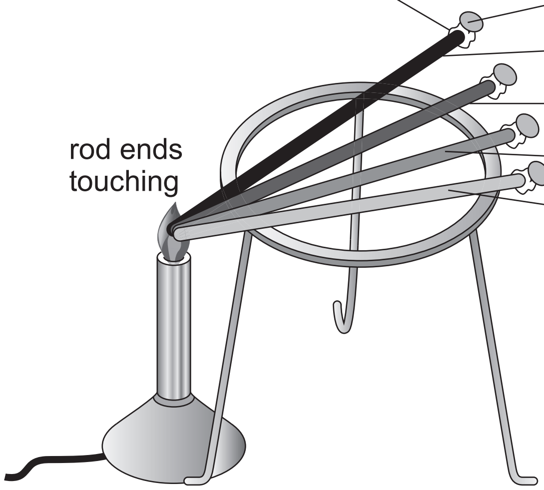
Examiner only

2. Aluminium, copper and iron are metals. Brass is a metal alloy. These materials have many uses.

The thermal conductivity of metals can be investigated using the apparatus below. The time taken for a drawing pin to fall from each metal rod is measured.



(a) (i) State the independent variable in this experiment. [1]

....

(ii) State **two** variables that are controlled in this experiment. [2]

....

....

(iii) Describe how the thermal conductivity is related to the dependent variable. [2]

....

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(b) Two students obtained the results shown in the table below.

Metal	Time taken for pin to drop (s)
iron	17
copper	7
brass	12
aluminium	5

(i) Use the results shown in the table above to place iron, copper, brass and aluminium in order of conductivity. [1]

Best conductor ....

....

....

Poorest conductor ....

(ii) The following table shows the theoretical values of the thermal conductivity of the metals.

Metal	Thermal conductivity (units)
iron	80.4
copper	401
brass	109
aluminium	210

Use the information above to explain whether the theoretical data agrees with experimental data. [3]

Best conductor ....

....

....

Poorest conductor ....

....

....

....

(c) Mercury is a metal which has a melting point of –39 °C. Suggest why the students could not use mercury in this investigation. [1]

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